

Long-Only Equity Capacity Methodology

Distributional Liquidity Migration, Execution Horizons and Alpha Decay

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Abstract

This document defines the stable methodology for estimating the capacity of a long-only equity strategy. It separates the methodology from portfolio-specific runs. Each run should produce a separate report containing the supplied inputs, fitted parameters, diagnostics, scenarios, thresholds and warnings. Appendices A–C define the user input contract and distinguish supplied, policy and fitted parameters.

Contents

1	Framework overview	2
2	Strategy economics	2
3	Distributional liquidity migration	2
4	Impact model	3
5	Execution horizon	3
6	Alpha decay	3
7	Net capacity	3
8	Required run-report contents	5
9	Validation requirements	5
A	User-supplied data inputs	5
B	User-set policy and scenario parameters	6
C	Fitted parameters and outputs	7

*with the help of ChatGTP

1 Framework overview

The framework maps assets under management (AUM) into a distribution of parent-order participation, execution horizons, implementation costs and captured alpha:

$$A \rightarrow F_A(p) \rightarrow D(p) \rightarrow \{\bar{c}(A), R_g(A)\} \rightarrow R_{\text{net}}(A), IR_{\text{net}}(A). \quad (1.1)$$

Turnover T is one-way annual turnover, so executed purchases and sales are approximated by $2TA$. Cost is expressed in basis points per executed notional.

The model consumes either aggregate ADV buckets or, preferably, parent-order data. Aggregate bucket shares are interpreted as shares of executed notional, not trade-count frequencies.

2 Strategy economics

At current AUM A_0 , the key quantities are gross relative return R_g , tracking error TE , turnover T and average holdings N :

$$IR_g = \frac{R_g}{TE}, \quad \alpha_{\text{holding}} = \frac{R_g}{N}, \quad \alpha_{\text{cycle}} = \frac{R_g}{T}, \quad H \approx \frac{1}{T}. \quad (2.1)$$

Current gross alpha is treated as already reflecting the current implementation schedule. Alpha decay at alternative AUM levels is therefore measured relative to the current schedule.

3 Distributional liquidity migration

Capacity depends on parent-order participation $p = Q/\text{ADV}$, not ADV in isolation. Modelling ADV alone would require a joint model for parent-order size and its dependence on ADV.

The aggregate implementation fits a Burr XII distribution:

$$F(p; c, d, \lambda) = 1 - \left[1 + \left(\frac{p}{\lambda} \right)^c \right]^{-d}. \quad (3.1)$$

It offers a flexible body and Pareto-like survival tail $1 - F(p) \sim p^{-cd}$. Empirical work finds power-law tails in trade-size distributions [1]. The calibration constrains $cd \geq 1.25$, preserving a heavy tail while ensuring finite mean participation.

The distribution changes with AUM through

$$\lambda(A) = \lambda_0 \left(\frac{A}{A_0} \right)^\eta, \quad d(A) = d_0 \left(\frac{A}{A_0} \right)^{-\kappa}. \quad (3.2)$$

The scale elasticity η captures how participation grows with AUM. Tail drift κ is reserved for structural changes in liquidity composition. Bucket shares follow smoothly from

$$w_i(A) = F_A(u_i) - F_A(\ell_i). \quad (3.3)$$

The participation distribution enters the capacity calculation through its quantile representation. The distribution is weighted by executed notional: each equal-width quantile interval represents an equal fraction of traded value, not an equal number of trades. At each AUM, set $F_A(p) = F(p; c, d(A), \lambda(A))$. Its inverse turns a fixed rank $q \in (0, 1)$ into participation:

$$p_A(q) = F_A^{-1}(q) = \lambda(A) \left[(1 - q)^{-1/d(A)} - 1 \right]^{1/c}. \quad (3.4)$$

The same rank is used at current and alternative AUM. This is a rank-preserving scenario assumption, not an observation that the same individual orders recur. It supplies the paired participations needed for the horizon ratio in equation (5.2). When $\kappa = 0$, $p_A(q) = p_{A_0}(q)(A/A_0)^\eta$; with tail drift, the full inverse in equation (3.4) is required.

Bucket weights in equation (3.3) summarize this distribution for charts. The continuous model does not multiply them into its costs a second time: the notional weights are already represented by the quantile average made explicit in Section 7.

4 Impact model

Expected and realised costs are fitted separately using

$$c(p) = a + bp^\gamma, \quad (4.1)$$

subject to positive parameter bounds. Square-root impact is an important benchmark, while the implementation permits the exponent to be fitted [2]. At current AUM, the continuous distributional expectation is calibrated back to the observed notional-weighted bucket cost.

5 Execution horizon

For target daily participation ρ and maximum horizon $D_{\max}$,

$$D(p) = \min\left(D_{\max}, \max\left(1, \left\lceil \frac{p}{\rho} \right\rceil\right)\right). \quad (5.1)$$

Current costs already embed the current schedule, so counterfactual effective participation is

$$p_{\text{eff},j}(A) = p_j(A) \frac{D_j(A_0)}{D_j(A)}. \quad (5.2)$$

The rule exposes the cost-versus-delay trade-off studied in optimal execution [3]. A production version can replace it with direct optimisation.

6 Alpha decay

Let H_α be signal half-life in trading days and $\delta = \log(2)/H_\alpha$. For equal daily slices over D days, average alpha capture is

$$g(D, H_\alpha) = \frac{1}{D} \sum_{t=0}^{D-1} e^{-\delta t} = \frac{1 - e^{-\delta D}}{D(1 - e^{-\delta})}. \quad (6.1)$$

Normalising to current implementation gives

$$R_g(A) = R_g(A_0) \mathbb{E} \left[\frac{g(D_j(A), H_{\alpha,j})}{g(D_j(A_0), H_{\alpha,j})} \right]. \quad (6.2)$$

Aggregate runs currently weight alpha by executed notional. Trade-level runs should weight by each order's forecast alpha contribution. Predictors with different mean-reversion speeds imply different desired trading speeds [4].

7 Net capacity

The participation distribution specified in Section 3 determines $p_A(q)$ through equations (3.1), (3.2) and (3.4). For each notional rank q , the execution rule in equation (5.1) is evaluated at current and alternative AUM:

$$\begin{aligned} D_A(q) &= D(p_A(q)), & D_0(q) &= D(p_{A_0}(q)), \\ p_{\text{eff},A}(q) &= p_A(q) \frac{D_0(q)}{D_A(q)}. \end{aligned} \quad (7.1)$$

Thus distributional migration has two channels: it changes effective participation and therefore impact, and it changes execution days and therefore alpha capture.

Let $\bar{c}_{\text{obs},0} = \sum_i w_{i,0}^{\text{obs}} c_i^{\text{obs}}$ be the observed current notional-weighted cost in basis points. The impact function $c(p)$ in equation (4.1) is the raw fitted curve. Its baseline calibration multiplier is

$$K = \frac{\bar{c}_{\text{obs},0}}{\int_0^1 c(p_{A_0}(q)) dq}. \quad (7.2)$$

Expected and realised curves each use their own multiplier. Making the expectation explicit, the expected cost and captured gross alpha are

$$\bar{c}(A) = K \int_0^1 c(p_{\text{eff},A}(q)) dq, \quad (7.3)$$

$$R_g(A) = R_g(A_0) \int_0^1 \frac{g(D_A(q), H_\alpha)}{g(D_0(q), H_\alpha)} dq. \quad (7.4)$$

Here g is equation (6.1), and a common half-life is used for the aggregate model. These integrals are the notional-weighted expectations previously written as $\mathbb{E}[\cdot]$. Integrating over uniform ranks is equivalent to integrating over the participation distribution supplied by Section 3, with current and alternative horizons paired at the same rank.

The expectations are evaluated numerically using J equally weighted midpoint quantiles:

$$\begin{aligned} q_j &= \frac{j - \frac{1}{2}}{J}, \quad j = 1, \dots, J, & K_J &= \frac{\bar{c}_{\text{obs},0}}{J^{-1} \sum_{j=1}^J c(p_{A_0}(q_j))}, \\ \bar{c}_J(A) &= \frac{K_J}{J} \sum_{j=1}^J c(p_{\text{eff},A}(q_j)), \\ R_{g,J}(A) &= \frac{R_g(A_0)}{J} \sum_{j=1}^J \frac{g(D_A(q_j), H_\alpha)}{g(D_0(q_j), H_\alpha)}. \end{aligned} \quad (7.5)$$

The factor $1/J$ supplies the notional weight. There is no additional bucket-share multiplier. Using the same numerical grid in the calibration denominator and cost average ensures $\bar{c}_J(A_0) = \bar{c}_{\text{obs},0}$ and $R_{g,J}(A_0) = R_g(A_0)$. The continuous impact expectation requires $cd(A) > \gamma$ for the Burr tail; the finite-mean participation constraint alone does not guarantee this for every fitted exponent or AUM. Finite-grid runs should check sensitivity to J , particularly under tail thickening.

The resulting impact estimate is converted from cost per executed notional into annual cost per unit of portfolio capital using one-way turnover:

$$C(A) = \frac{2T\bar{c}(A)}{10,000}. \quad (7.6)$$

The conversion by 10,000 changes basis points into a decimal return. Net alpha and IR are

$$R_{\text{net}}(A) = R_g(A) - C(A), \quad IR_{\text{net}}(A) = \frac{R_{\text{net}}(A)}{TE}. \quad (7.7)$$

The quantities R_g , C and TE are expressed as decimal annual returns, and T as decimal one-way turnover. Distributional migration therefore affects both components of net return: impact in $C(A)$ and delay in $R_g(A)$. Tracking error and turnover remain fixed assumptions in this specification.

For example, consider a notional rank whose participation increases from 8% to 16% ADV. With a 10% daily target and a non-binding horizon cap, its horizon increases from one to two days. Effective participation becomes $16\% \times 1/2 = 8\%$, so its modeled impact is unchanged, while its alpha-capture ratio becomes $g(2, H_\alpha)/g(1, H_\alpha) < 1$. Migration can therefore lower net IR through delay even when scheduling offsets the impact increase. This is an illustrative rank, not a claim about the entire portfolio.

A run report should show the complete capacity curve and explicitly selected thresholds rather than presenting a single universal capacity number.

8 Required run-report contents

Every portfolio run report must include:

1. run identity, timestamp, code version and input source;
2. supplied strategy, liquidity, execution and alpha-decay inputs;
3. fitted distribution and impact parameters with fit diagnostics;
4. current-AUM reconciliation to observed costs and alpha;
5. AUM scenario table and continuous capacity curves;
6. bucket migration and execution-horizon diagnostics;
7. selected capacity thresholds and exact crossing estimates;
8. data-quality, extrapolation and model-risk warnings.

9 Validation requirements

A valid run should confirm that notional shares sum to one, AUM and half-life are positive, participation buckets are ordered and non-overlapping, expected and realised costs share a common benchmark, and current-AUM impact reconciles to the supplied weighted cost. The report should flag missing trade-level alpha weights, large extrapolations, horizons at $D_{\max}$, unstable tail fits and sensitivity to user policy parameters.

A User-supplied data inputs

Table 1: Required and optional user data

Input	Requirement	Unit / format	Definition and validation
Current AUM A_0	Required	Currency bn	Average deployed AUM represented by the liquidity sample; must be positive.
Gross expected alpha R_g	Required	Decimal p.a.	Expected relative return before modeled impact; document whether existing live slippage is embedded.
Tracking error TE	Required	Decimal p.a.	Annualised ex-ante or realised relative volatility; must be positive and methodologically aligned with alpha.
One-way turnover T	Required	Decimal p.a.	Purchases or sales divided by average AUM. Do not supply two-way turnover without changing the convention.
Average holdings N	Diagnostic	Count	Average number of portfolio positions over the measurement period.
Bucket lower and upper bounds	Required for aggregate run	Fraction of ADV	Ordered, non-overlapping participation intervals covering the full positive range.
Executed-notional share	Required for aggregate run	Fraction summing to 1	Annual executed-notional mass in each bucket; not trade-count frequency.
Representative participation	Required for impact fit	Fraction of ADV	Prefer the notional-weighted mean participation within each bucket.
Expected impact	Required	bp / executed notional	Expected implementation cost using a consistent arrival-price or other declared benchmark.

Input	Requirement	Unit / format	Definition and validation
Realised impact	Recommended	bp / executed notional	Realised cost using the same benchmark as expected impact.
Parent-order notional or shares	Required for trade-level run	Currency / shares	Parent order before child-order slicing, net of cancellations according to a declared convention.
Security ADV	Required for trade-level run	Currency shares/day or	ADV measured consistently with order units, including declared lookback and lag.
Order timestamps	Recommended	Datetime	Decision, release, first fill and final fill times.
Actual execution horizon	Recommended	Trading days	Number of sessions over which the parent order was worked.
Forecast alpha	Recommended	bp or return	Expected contribution at the order decision timestamp.
Signal family	Recommended	Category	Used to assign or estimate signal-specific decay.
Observed alpha path	Preferred	Return by horizon	Forward alpha after 1, 2, 3, 5, 10 or more trading days for empirical half-life estimation.
Side, spread and volatility	Recommended	Category / bp / decimal	Used for richer asymmetric impact and execution-risk models.

B User-set policy and scenario parameters

Table 2: Parameters chosen or approved by the user

Parameter	Example	Unit	Interpretation
Participation-scale elasticity η	0.85	Elasticity	Response of the Burr scale to AUM. Set to 1 for proportional scaling; estimate from portfolio rescaling where possible.
Tail-thickening elasticity κ	0.00	Elasticity	Structural change in upper-tail liquidity composition. Keep at zero unless supported by portfolio simulations or observations.
Target daily participation ρ	10%	ADV/day	Execution-policy target used to determine horizon.
Maximum execution horizon $D_{\max}$	10	Trading days	Maximum permitted horizon before daily participation exceeds the target.
Alpha half-life H_{α}	10	Trading days	Time for forecast alpha to halve. Prefer signal-specific empirical estimates.
AUM scenario grid	1–100	Currency bn	Range over which curves and thresholds are evaluated.
Net IR threshold	0.75	Ratio	Minimum acceptable net information ratio.
Alpha-retention threshold	80%	Percentage	Minimum fraction of current gross alpha remaining after delay and impact.
Impact-drag threshold	100	bp p.a.	Maximum acceptable annual return drag.
Alpha-capture threshold	95%	Percentage	Minimum gross alpha retained after execution delay, before impact.

C Fitted parameters and outputs

The user does not normally supply Burr parameters (c, d, λ_0) or impact parameters (a, b, γ) ; the model estimates them. Every run report should disclose them along with the tail index cd , bucket-fit residuals, impact-fit residuals, baseline calibration multiplier, threshold crossings and warnings. Parameters should be versioned with their input sample and estimation date.

References

- [1] R. Rak et al. “Tests of scaling and universality of the distributions of trade size and share volume.” *Physical Review E* 76, 046109 (2007). doi:10.1103/PhysRevE.76.046109.
- [2] J. D. Farmer et al. “How efficiency shapes market impact.” *Quantitative Finance* 13(11), 1743–1758 (2013). arXiv:1102.5457.
- [3] R. Almgren and N. Chriss. “Optimal Execution of Portfolio Transactions.” *Journal of Risk* 3, 5–39 (2001). doi:10.21314/JOR.2001.041.
- [4] N. Gârleanu and L. H. Pedersen. “Dynamic Trading with Predictable Returns and Transaction Costs.” *Journal of Finance* 68(6), 2309–2340 (2013).